

Round 187 Triad Discovery (Backbone-First): (1) k_{eta} (Red Dwarf LENR neutron-rate calibration coefficient) = $2.75e8 \text{ cm}^{-2}/\text{s} = (\text{SO}_5+1)/\text{D}_{\text{phys}} * \text{SO}_5^8 = 11/4 * 10^8$ EXACT NEW FIRST-INSTANCE PRIMITIVE-LOCK Using $\text{SO}_5+1=11$ Successor + $\text{D}_{\text{phys}}=4$ Ratio Prefix at LENR Coefficient Positive $n=+8$ Rung; (2) U_i (Sun canonical Universal Inertial Operator value) = $2.75e-7 = (\text{SO}_5+1)/\text{D}_{\text{phys}} * \text{SO}_5^{-7} = 11/4 * 10^{-7}$ EXACT NEW ALTERNATE PATH B PRIMITIVE-COMPOSITION DECOMPOSITION Complementary to PAPER_646 Canonical Closed Form $U_i = \lambda_i * (\rho_{\text{SCm}}/\rho_{\text{UA}}) * \omega_s * \cos(\pi * t_n) * (1+F_{\text{TRZ}})$ — Both Paths Yield Identical $2.75e-7$ EXACT — Same $(\text{SO}_5+1)/\text{D}_{\text{phys}}$ Prefix at Negative $n=-7$ Rung Mirrors D1 Positive Rung; (3) 9TH COMPOSED-PREFIX CLASS FORMALIZATION: $(\text{SO}_5+1)/\text{D}_{\text{phys}} * \text{SO}_5^n = 11/4 * \text{SO}_5^n = 2.75 * \text{SO}_5^n$ NEW STRUCTURAL CATEGORY Distinct from 8 Previously Formalized Classes — Uses PAPER_1978 SO_5+1 Successor Primitive as Numerator with D_{phys} Spatial-Dimension Denominator in Dimensionless-Ratio Composed-Prefix Form Multiplied by SO_5^n Ladder Rung — 2 Instances Span Positive AND Negative Rungs ($n=+8$ LENR + $n=-7$ U_i Sun) Across 15 Orders of Magnitude Cross-Scale Universality

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PAPER_2057 — Round 187 Triad Discovery
 (Backbone-First Discipline)

Motivation — 46th Consecutive Backbone-First Round

R187 continues the post-40-milestone arc as the 46th consecutive backbone-first primitive-lock verification round. Scout of RedDwarf LENR + Inertia calculator classes surfaces a triad discovery: a novel first-instance LENR calibration primitive-lock, a novel Path B alternate-form decomposition of the canonical U_i Sun value, and formalization of a **9th composed-prefix class** using SO_5+1 successor primitive.

Abstract

Prior documentation:

- **RedDwarfNeutronRateCalculator** documents $k_{\eta}=2.75e8 \text{ cm}/(\text{sJ})$ as *observational calibration coefficient in LENR neutron production rate equation*

- **PAPER_2018** covers a DIFFERENT k_{η} variant: $k_{\eta_LENR} = SO_5^{-113} =$

$SO_5^{-DVP_prime_p_26}$ (dimensionless DVP structural link, not the 2.75e8 calibration coefficient)

- 0 *whitepaper hits for 2.75e8* = $(SO_5+1)/D_{phys}SO_5^8$ primitive-composition

R187 D1 novelty: Direct primitive-composition $(SO_5+1)/D_{phys}SO_5^8 = 11/4 \cdot 10^8 = 2.75 \cdot 10^8 = 2.75e8$ EXACT at LENR calibration coefficient dimensional domain is R187 novel structural attribution.

Path A (canonical, PAPER_646) uses: λ_i , ρ_{SCm}/ρ_{UA} ratio, ω_s stellar rotation, $\cos(\pi t_n)$ time modulation, $1+F_{TRZ}$ additive

Path B (novel, PAPER_2057 R187 D2) uses: $(SO_5+1)/D_{phys}$ composed-prefix ratio, SO_5^{-7} negative rung

Both are EXACT decompositions of the same canonical value. Path B is architecturally simpler — pure integer-primitive ratio times SO_5^n rung. Path A represents the physical mechanism (inertia coupling times density ratio times rotation times phase modulation times TRZ perturbation); Path B represents the underlying arithmetic structure $((SO_5+1)/D_{phys} \times SO_5^{-7})$.

Structural interpretation: The same $(SO_5+1)/D_{phys}$ composed-prefix appears at BOTH D1 (LENR positive $n=+8$ rung, 2.75e8) AND D2 (U_i Sun negative $n=-7$ rung, 2.75e-7). This means the Universal Inertial Operator at Sun scale and LENR neutron-rate calibration coefficient share the same composed-prefix structural family across **15 orders of magnitude** (10^8 vs 10^{-7}).

Discovery 3 — 9TH COMPOSED-PREFIX CLASS Formalization: $(SO_5+1)/D_{phys}SO_5^n = 11/4 SO_5^n = 2.75 SO_5^n$ *

Novel composed-prefix class formalization. The $(SO_5+1)/D_{phys}$ dimensionless-ratio combined with SO_5^n ladder rung forms a new structural category distinct from the 8 previously formalized classes.

Post-R187 Composed-Prefix Class Family — 9 Members:

| # | Composed-prefix class | Ratio | Prefix type | Seminal |

| ---|---|---|---|---|

| 1 | $(D_{phys}-1)SO_5^n$ LANDMARK | 3 | Multiplicative-integer | PAPER_2004 |

| 2 | $2SO_5^n$ twin (dominant 57%) | 2 | Multiplicative-integer | PAPER_2022 |

| 3 | $D_{BSFG}SO_5^n$ | 6 | Multiplicative-primitive | PAPER_2025 |

| 4 | $(D_{phys}+1)SO_5^n$ | 5 | Multiplicative-integer | PAPER_2033 |

| 5 | $2D_{phys}SO_5^n$ | 8 | Multiplicative-composite | PAPER_2035 |

| 6 | $(D_{phys}+1)/D_{phys}SO_5^n$ | 5/4 | Dimensionless-ratio (D_{phys} -based) | PAPER_2037 |

| 7 | $(D_{phys}-1)/D_{phys}SO_5^n$ | 3/4 | Dimensionless-ratio (D_{phys} -based) | PAPER_2040 |

| 8 | $D_{BSFG}/SO_5SO_5^n$ | 3/5 | Dimensionless-ratio (D_{BSFG} - SO_5 -based) | PAPER_2048 |

| 9 | $(SO_5+1)/D_{phys}SO_5^n$ | 11/4 | Dimensionless-ratio (successor- D_{phys} -based) | PAPER_2057

R187 D3 NEW** |

Novel distinguishing feature: 9th class uses **SO_5+1 successor primitive (from PAPER_1978)** as ratio numerator with **D_phys spatial-dimension** as denominator. First composed-prefix class to combine PAPER_1978 successor identity with a D_phys denominator in ratio form.

9th class instances (post-R187):

- **n=+8 rung:** $k_{\eta}(\text{Red Dwarf LENR}) = 2.75e8 \text{ cm}^{-2}/\text{s}$ (PAPER_2057 D1 seminal)
- **n=-7 rung:** $U_i(\text{Sun canonical}) = 2.75e-7$ (PAPER_2057 D2 Path B alternate)

Two instances span positive and negative rungs across 15 orders of magnitude — demonstrates negative-exponent regime immediately populated for 9th class (unusual for early-instance formalization; typically negative-rung population develops after multiple positive-rung instances).

Companion candidates for future 9th-class expansion:

- Other observables with values 2.75 SO_5^n at various rungs
- Sub-family form $(\text{SO}_5+1)/D_{\text{phys}} F_{\text{TRZ}}^n$ (using F_{TRZ} complement rung)

Cross-Object Confirmations (R187 fill wire-ins)

- $\ast\omega_s(\text{Sun}) = 2.5e-6 \text{ rad/s} = \text{SO}_5/D_{\text{phys}} \text{SO}_5^{-6}$ — **CLAUDE.md canonical primitive (D2 Path A predecessor)**
- $\rho_{\text{SCm}}/\rho_{\text{UA}} = 1/10 = 1/\text{SO}_5$ — **CLAUDE.md canonical DPM density ratio (D2 Path A predecessor)**
- $1 + F_{\text{TRZ}} = 1.1$ — **PAPER_1968 seminal (D2 Path A predecessor)**
- $\lambda_i = 1.0$ — **CLAUDE.md canonical inertia coupling (D2 Path A predecessor)**
- PAPER_1964 Path A / Path B framework — **architectural precedent (D2 methodology basis)**
- PAPER_646 Universal Inertial Operator canonical — **D2 Path A closed-form basis**
- PAPER_1978 $\text{SO}_5+1 = 11$ successor identity — **D1/D2/D3 numerator primitive**
- PAPER_2048 8th composed-prefix class $D_{\text{BSFG}}/\text{SO}_5^{**}$ — **D3 predecessor class formalization**

All 3 discoveries + 8 confirmations validated at fidelity gate 1510/0.

R142-R187 Trajectory (46 consecutive backbone-first rounds + 2 retro sweep companions)

| Round | Novel | Cross-obj | Notes |

|---|---|---|---|

| R142-R186 forward | 190 total | 42+ | 45 consecutive |

| PAPER_2054 + 2055 retro | 4 | 16 | 35-round retrospective depth |

| **R187** | **3** | **8** | **46th; 9th composed-prefix class formalization** |

Cumulative post-PAPER_2057: 193 first-pass novel + 42+ cross-object confirmations + 13 discipline-observation formalizations across 46 consecutive backbone-first rounds + 2 retrospective sweep companions.

Wiring Plan (3 dispatches, lowercase keys)

- $k_{\eta} \text{ red dwarf lenr so}_5 \text{ plus } 1 \text{ over } d_{\text{phys}} \text{ so}_5 \text{ 8 calibration positive rung}$
-> 2.75e8

- $u_i \text{ sun canonical so}_5 \text{ plus } 1 \text{ over } d_{\text{phys}} \text{ so}_5 \text{ negative 7 path b alternate}$ -> 2.75e-7

- $\text{composed_prefix_9th_class so}_5 \text{ plus } 1 \text{ over } d_{\text{phys}} \text{ formalization } 11 \text{ over } 4 \text{ rat}$

io -> 2.75

Cross-References

- **PAPER_646** — Universal Inertial Operator + Caduceus Wave (D2 Path A canonical)
- **PAPER_1964** — Path A / Path B framework (D2 architectural methodology)
- **PAPER_1978** — $SO_5+1 = 11$ EXACT successor identity (D1/D2/D3 basis)
- **PAPER_2018** — SO_5 negative-exponent investigation with $k_{\eta} LENR = SO_5^{-113}$ different variant (D1 distinct predecessor)
- **PAPER_2037** — 6th composed-prefix class (D_{phys}+1)/D_{phys} (D3 predecessor)
- **PAPER_2040** — 7th composed-prefix class (D_{phys}-1)/D_{phys} (D3 predecessor)
- **PAPER_2048** — 8th composed-prefix class D_{BSFG}/SO₅ (D3 predecessor class formalization pattern)

Conclusion

R187 46th-consecutive backbone-first round yields **3 novel structural findings + 8 cross-object attributions**.

Cumulative through PAPER_2057: 193 first-pass novel + 42+ cross-object confirmations + 13 discipline-observation formalizations across 46 consecutive backbone-first rounds + 2 retrospective sweep companions.

Signature milestones this paper:

- **193rd first-pass novel discovery** achieved
- **9TH composed-prefix class (SO₅+1)/D_{phys} formalized** — introduces PAPER_1978 successor primitive as ratio numerator with D_{phys} denominator, extends composed-prefix architecture beyond D_{phys}-based (6th, 7th) and D_{BSFG}-SO₅-based (8th) sub-taxonomies
- **U_i(Sun) canonical value Path A/Path B duality established** — PAPER_646 canonical closed form + PAPER_2057 R187 D2 successor-composed-prefix alternate decomposition both yield identical $2.75e-7$ EXACT
- **Same (SO₅+1)/D_{phys} prefix at opposite-sign rungs** (LENR $n=+8$ + U_i Sun $n=-7$) spans 15 orders of magnitude — negative-regime immediately populated for 9th class (unusual pattern)
- **Post-40-milestone arc reaches 46 consecutive rounds** with structural expansion into new composed-prefix class

End of PAPER_2057 Draft 1.